AquaWatch: A Smart Water Monitoring and Conservation System

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I. Introduction

Water is necessary for drinking, sanitation, agriculture, health, and daily activities. However, many communities face water scarcity because the demand for clean water can exceed the available supply. Population growth, climate change, pollution, overuse of water resources, and poor water management can increase this problem (Biswas et al., 2025; Salehi, 2022).

Water scarcity affects both developing and developed countries. It can limit access to safe water and affect health, education, agriculture, and livelihoods. Caritas Australia reports that more than 2.2 billion people worldwide do not have access to safe drinking water (Caritas Australia, n.d.).

The Philippines also faces water supply problems. Palanca-Tan (2020) reported that 9 million out of 101 million Filipinos at the time of the study relied on unimproved, unsafe, or unsustainable water sources. The study also identified inadequate water infrastructure and poor management as important factors affecting water access in the country.

Technology can help improve how people monitor and manage water. Water monitoring systems can collect information about water use, identify possible water losses, and help users make better decisions. Tzanakakis et al. (2020) identified improved water management, water-use efficiency, and technological solutions as important approaches to addressing water scarcity.

This project proposes AquaWatch, a smart water monitoring and conservation system. The system aims to help households and institutions monitor water consumption, identify possible leaks, and reduce unnecessary water use.

II. Problem Description

Water scarcity has several causes. Climate change can affect rainfall and increase the risk of drought. Population growth increases the demand for water. Overuse of groundwater and other water resources can reduce available supplies. Poor infrastructure and water management can also contribute to water shortages (Biswas et al., 2025; Salehi, 2022).

Water scarcity can affect people's daily lives. Limited water access can make drinking, cooking, hygiene, and sanitation more difficult. It can also affect agriculture and food production because farming requires a large amount of water. The effects can become more serious in communities that already have limited access to water infrastructure (Caritas Australia, n.d.).

The problem is also present in the Philippines. Palanca-Tan (2020) found that some Filipino households experienced inadequate and intermittent water supplies. The study also linked water problems to insufficient infrastructure and inefficiencies in water management.

Another concern is water wastage. Leaking pipes, faucets, and other water systems can cause unnecessary water loss. Users may not notice these problems immediately. They may also have limited information about how much water they consume each day.

A lack of monitoring can make water management more difficult. Tzanakakis et al. (2020) emphasized the importance of improved water management, monitoring, and technologies that increase water-use efficiency.

These problems show the need for a simple system that can help users monitor their water use. Technology can provide users with real-time information and alerts that can support better water management.

III. Proposal Solution

The proposed solution is AquaWatch, a smart water monitoring and conservation system. It uses water flow sensors, internet connectivity, and a mobile or web application to monitor water consumption. The system measures the amount of water passing through a pipe. A water flow sensor collects the data and sends it to the AquaWatch system. Users can then view their water consumption through a digital dashboard.

AquaWatch will have four main features.

First, the system will provide water consumption monitoring. Users can view their daily, weekly, and monthly water use. This allows them to understand their consumption patterns.

Second, the system will provide leak detection. If the sensor detects continuous or unusually high water flow, AquaWatch will send an alert. Users can then check their pipes, faucets, or other water sources for possible leaks.

Third, the system will provide water-saving reminders. Users can receive reminders to turn off unused faucets, repair leaks, and reduce unnecessary water consumption.

Fourth, the system will provide water-use reports. Users can compare their consumption over time and set simple water-saving goals.

The system may use a water flow sensor, Arduino or ESP32 microcontroller, Wi-Fi connection, cloud storage, and a mobile or web application.

The basic process is:

Water Flow → Sensor → Internet → AquaWatch Application → Alert → User Action

For example, a household may normally have very little water use during the night. If AquaWatch detects continuous water flow for several hours, the system can notify the household. The user can then check for a leaking faucet or pipe.

The system can also be used in schools. A school can monitor water consumption in different areas and identify unusual usage. This information can help administrators identify possible leaks and encourage students and staff to conserve water.

AquaWatch does not aim to create additional water. Instead, it focuses on improving how available water is monitored and used. This approach supports research that recommends better water management and technologies that improve water-use efficiency (Tzanakakis et al., 2020).

The project can also support existing water conservation strategies. Research on water scarcity highlights the importance of efficient water use and technological improvements in managing limited water resources (Biswas et al., 2025).

IV. Expected Benefits

AquaWatch can provide several benefits.

For households, it can help users understand their water consumption and identify possible leaks.

For schools, it can help monitor water use and encourage students and staff to practice conservation.

For other institutions, the system can provide information that can support water management decisions.

The main expected benefits are reduced water waste, earlier detection of possible leaks, greater awareness of water consumption, and improved water-saving habits.

V. Conclusion

Water scarcity affects communities around the world, including the Philippines. Climate change, population growth, overuse of water resources, poor infrastructure, and inefficient management can increase pressure on available water supplies (Biswas et al., 2025; Palanca-Tan, 2020; Salehi, 2022).

AquaWatch provides a technology-based approach to this problem. The system allows users to monitor water consumption, identify unusual water flow, receive alerts, and track their water-use patterns.

The proposed system focuses on a practical part of water conservation. It helps users recognize water waste and respond to possible leaks. By combining water flow sensors, internet connectivity, and a digital application, AquaWatch can support more efficient water use in households, schools, and other institutions.

Technology alone cannot solve water scarcity. However, better monitoring and responsible water use can help reduce unnecessary water loss. AquaWatch demonstrates how information technology can support practical water conservation efforts.

References

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